IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 5 October 2025, accepted 12 November 2025, date of publication 18 November 2025, date of current version 24 November 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3634156

RESEARCH ARTICLE

Superpixel-Guided Graph-Attention Boundary GAN for Adaptive Feature Refinement in Scribble-Supervised Medical Image Segmentation

MANSOOR HAYAT [1], (Member, IEEE), AND SUPAVADEE ARAMVITH [2], (Senior Member, IEEE)
[1]Section of Neurosurgery, Department of Surgery, Rady Faculty of Health Sciences, University of Manitoba, Winnipeg, MB R3T 5V6, Canada
[2]Multimedia Data Analytics and Processing Research Unit, Department of Electrical Engineering, Faculty of Engineering, Chulalongkorn University, Bangkok 10330, Thailand

Corresponding author: Supavadee Aramvith (supavadee.a@chula.ac.th)

This work was supported by Thailand Science Research and Innovation Fund Chulalongkorn University under Grant IND_FF_69_105_2100_018.

ABSTRACT Fully supervised medical image segmentation still relies on labor-intensive, pixel-level annotations, which limits scale across cohorts and imaging settings. Scribble supervision reduces this burden, yet many CNN-based methods struggle under sparse labels due to weak global context and poor boundary handling. We address these issues with **SGGAB-GAN**, a scribble-supervised framework that uses adversarial learning, residual attention, and an enhanced feature pipeline built upon two modules: the *Superpixel-Guided Graph-Attention Boundary (SGGAB) block* and the *Adaptive Feature Refinement Block (AFRB)*. First, the *SGGAB* block propagates limited scribble cues over a superpixel graph and reinjects boundary information, yielding crisp edges even with few annotations. Second, the *AFRB* fuses global context with local detail and works with residual attention gates to focus on anatomically relevant regions. On ACDC and MSCMRseg, SGGAB-GAN attains average Dice scores of 0.902 and 0.871, respectively, outperforming scribble-based methods such as ScribFormer and CycleMix while narrowing the gap to full supervision to under 2%. These results indicate that SGGAB-GAN provides high-quality segmentation at a fraction of the labeling cost, making it a scalable choice.

INDEX TERMS Scribble supervision, weakly supervised medical segmentation, superpixel-guided graph attention, generative adversarial networks, adaptive feature refinement.

I. INTRODUCTION

Medical image segmentation is essential for numerous clinical applications, such as disease diagnosis, treatment planning, and patient monitoring. Recently, convolutional neural networks (CNNs), particularly architectures such as U-Net, have become the standard method due to their excellent performance in accurately segmenting anatomical structures from medical images [1], [2]. However, CNN-based segmentation models typically require extensive fully annotated datasets, where each pixel of the anatomical regions of interest must be labeled precisely. The manual labeling of medical images is labor-intensive, time-consuming, and costly, as it involves significant effort from expert clinicians [3].

To overcome the reliance on fully annotated datasets, researchers have explored weakly-supervised segmentation methods that use less detailed forms of annotations such as bounding boxes, points, and scribbles [3]. Among these approaches, scribble annotations have drawn substantial attention due to their simplicity and efficiency, providing a convenient balance between annotation effort and the information content required for effective segmentation [4]. However, existing CNN-based methods face critical limitations

The associate editor coordinating the review of this manuscript and approving it for publication was Zhan-Li Sun.

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

when using scribble annotations due to insufficient contextual information, making it challenging to capture accurate boundaries and subtle anatomical features [5].

Previous methods for scribble-supervised segmentation can be categorized mainly into two approaches. The first approach expands sparse scribble annotations into dense labels using iterative label refinement techniques or conditional random fields (CRFs). While effective, this strategy often introduces noisy pseudo-labels that negatively affect segmentation accuracy [6], [7]. The second approach utilizes adversarial training frameworks, employing additional fully annotated masks to learn global shape priors. Although these methods improve segmentation quality, the requirement of extra annotations undermines the primary advantage of weak supervision, which aims to minimize annotation effort [4], [8].

Recent advances such as hybrid CNN-Transformer models, notably ScribFormer, have shown promising improvements by integrating global context and detailed local information [5]. Despite these advancements, such hybrid models still have challenges in effectively integrating local structural details with global context due to limitations in their feature refinement mechanisms.

To address the shortcomings of existing scribble-supervised approaches, we present **SGGAB-GAN**, an adversarial segmentation framework that augments our residual-attention U-Net with two tightly coupled components. First, each skip connection is processed by a *Superpixel-Guided Graph-Attention Boundary (SGGAB) block*: the encoder feature map is over-segmented into superpixels, a region-adjacency graph is built, and graph attention diffuses the sparse scribble labels across neighbouring regions. The updated region features are then scattered back to pixels and fused with an edge map derived from scribble gradients, giving boundary-aware, context-rich skip features. Second, an *Adaptive Feature Refinement Block (AFRB)* follows every up-sampling step, blending these refined skips with decoder activations while residual attention units along the encoder–decoder paths keep the network focused on anatomically salient structures. A lightweight PatchGAN discriminator further encourages realistic mask topology under limited supervision.

Experiments on the ACDC [10] and MSCMRseg [11] cardiac MRI benchmarks confirm the effectiveness of this design. **SGGAB-GAN** attains average Dice scores of **0.902** and **0.871**, respectively, surpassing state-of-the-art scribble-supervised methods such as ScribFormer [5] and CycleMix [6], and closing the gap to fully supervised baselines to within two percentage points. Ablation studies show that SGGAB contributes the largest individual gain, while AFRB and residual attention provide complementary improvements and stabilise training.

Our main contributions are:

- SGGAB-GAN, GAN-based scribble supervised medical segmentation framework that integrates superpixel-level graph attention and boundary cues into a scribble-supervised setting.
- SGGAB block, which propagates sparse scribble information over a superpixel graph and reinjects edge detail, yielding crisp and coherent masks.
- We refine decoder features with AFRBs and residual attention, achieving a balanced fusion of global context and local detail.
- Comprehensive evaluation on two public datasets demonstrates that SGGAB-GAN sets a new state-of-the-art for scribble-supervised medical image segmentation while approaching fully supervised performance at a fraction of the annotation cost.

II. LITERATURE REVIEW

A. LEARNING FROM SCRIBBLE SUPERVISION

Scribble annotations, characterized by sparse labeling where only a fraction of pixels are annotated, offer a cost-effective alternative to dense annotations in medical image segmentation. However, the limited supervision signal poses challenges in training effective segmentation models.

To address this, various methods have been proposed. Luo et al. [7] introduced a dual-branch network with one encoder and two decoders, dynamically mixing the decoders' predictions to generate pseudo labels for auxiliary supervision, effectively learning from scribble annotations. Qu et al. [12] proposed ScribSD+, a framework based on multi-scale knowledge distillation and class-wise contrastive regularization, significantly improving segmentation performance under scribble supervision. Additionally, Li et al. [13] developed ScribbleVC, a framework leveraging vision and class embeddings via a multimodal information enhancement mechanism, demonstrating improved accuracy and efficiency in segmentation tasks.

B. GENERATIVE ADVERSARIAL NETWORKS IN MEDICAL IMAGE SEGMENTATION

Generative Adversarial Networks (GANs), introduced by Goodfellow et al. [15] in 2014, have significantly impacted the field of medical image analysis, in image segmentation tasks as well. The unique adversarial framework of GANs, comprising a generator and a discriminator, enables the modeling of complex data distributions, facilitating the generation of realistic synthetic images. This capability has been harnessed to augment limited medical datasets, thereby enhancing the performance of segmentation models.

In medical image segmentation, GANs have been employed to address challenges such as data scarcity and the need for precise annotations. For instance, GAN-based architectures have been utilized to generate synthetic medical images, effectively expanding training datasets and addressing class imbalance issues. This augmentation strategy enhances the robustness and generalization capability of segmentation models.

Several studies have demonstrated the efficacy of GANs in medical image segmentation. For example, Han et al. [16]

utilized GANs to generate realistic brain MR images from latent Gaussian spaces, highlighting the potential of GANs in producing high-quality medical images. Frid-Adar et al. [17] employed GANs to generate synthetic liver lesion images, improving the performance of lesion classification algorithms. Additionally, Iqbal et al. [18] provided a comprehensive overview of GAN applications in medical imaging, underscoring their versatility in various tasks, including segmentation.

C. GANS FOR WEAKLY SUPERVISED MEDICAL IMAGE SEGMENTATION

Weakly supervised learning aims to reduce the dependency on extensive pixel-level annotations by utilizing weaker forms of supervision, such as image-level labels, bounding boxes, or scribbles. GANs have been effectively integrated into weakly supervised frameworks to improve segmentation performance under such constraints [19], [20].

1) ADVERSARIAL TRAINING WITH PARTIAL ANNOTATIONS

In scenarios with limited annotations, adversarial training has been utilized to guide segmentation networks. For instance, Hu et al. [20] proposed a weakly supervised method using generative adversarial training for nuclei segmentation, achieving improved performance despite the scarcity of labeled data. Similarly, Valvano et al. [21] introduced a multi-scale adversarial attention gate mechanism, leveraging GANs to learn from scribble annotations for medical image segmentation.

2) QUALITATIVE COMPARISON WITH EXISTING METHODS

3) GANS FOR DATA AUGMENTATION

GANs have also been employed to generate synthetic medical images, effectively expanding training datasets and addressing class imbalance issues. This augmentation strategy enhances the robustness and generalization capability of segmentation models. For example, Frid-Adar et al. [17] utilized GANs to generate synthetic liver lesion images, thereby improving the performance of lesion classification algorithms. Skandarani et al. [22] conducted an empirical study on GANs for medical image synthesis, demonstrating their effectiveness in generating high-quality synthetic data for augmentation purposes.

4) DOMAIN ADAPTATION WITH GANs

Domain adaptation techniques using GANs have been developed to address the variability in medical images arising from different imaging modalities or protocols. By aligning the distributions of source and target domains, GANs facilitate the transfer of knowledge from labeled datasets to unlabeled ones. Yan et al. [23] proposed a Unet-GAN framework for vendor adaptation in medical image segmentation, demonstrating improved cross-domain performance. Additionally, Mondal et al. [24] introduced a few-shot 3D multi-modal medical image segmentation approach using generative adversarial learning, highlighting the potential of GANs in domain adaptation scenarios.

5) GANs IN SCRIBBLE-SUPERVISED MEDICAL IMAGE ANALYSIS

Generative adversarial networks (GANs) have become a popular remedy for the sparse and noisy supervision characteristic of scribble labels. Mao et al. [25] first introduced WSGAN, where a discriminator enforces structural plausibility while a generator synthesises auxiliary images to compensate for annotation scarcity. Valvano et al. [21] later combined multi-scale adversarial attention gates with scribble masks, obtaining notable gains on cardiac-MR segmentation. More recent studies adopt patch-level critics: Lin et al. [31] constrain local shape consistency, whereas Zheng et al. [32] embed a CycleGAN backbone to bridge domain gaps under weak cues. These approaches, however, still suffer from label leakage across weak boundaries because pixel- or patch-wise critics do not explicitly follow object contours. Our framework alleviates this problem by aligning adversarial feedback with superpixel boundaries through the proposed Superpixel-Guided Graph Attention Boundary (SGGAB) module and by coupling this boundary-aware guidance with a masked Dice + CE objective and a lightweight PatchGAN critic, striking a balance between boundary precision and training stability while adding only 0.4 M parameters.

III. METHOD

Residual Attention U-Net (RAUNet) [26] has become a popular baseline for fully supervised segmentation; it employs a ResNet-34 encoder and an Augmented Attention Module (AAM) that performs channel-wise recalibration. Although powerful, RAUNet has two drawbacks in the weakly supervised setting: (i) the pre-trained backbone is heavy, which limits deployment on resource-constrained hardware, and (ii) global channel attention alone is insufficient when labels are sparse because the network must still infer fine boundaries from incomplete scribbles. Our **SGGAB-GAN** removes these shortcomings by introducing *boundary-aware residual blocks*, *spatial Attention Gates*, and two new purpose-built modules, namely the *Superpixel-Guided Graph-Attention Boundary (SGGAB) block* and the *Adaptive Feature Refinement Block (AFRB)*. The result is a lighter, faster, and more data-efficient architecture designed specifically for scribble supervision.

A. ENCODER: LIGHTWEIGHT RESIDUAL STAGES

1) **Residual units.** Each stage contains two consecutive 3×3 convolutions followed by Batch Normalisation and ReLU, wrapped in an identity skip connection so that the residual mapping $\mathcal{F}_i$ can be trained easily even with few annotations:

$$e_i = \mathrm{ReLU}\big(x_{i-1} + \mathcal{F}_i(x_{i-1})\big), \qquad x_0 = x. \tag{1}$$

2) **Progressive abstraction.** A 2×2 max-pool reduces spatial resolution after each residual unit, yielding

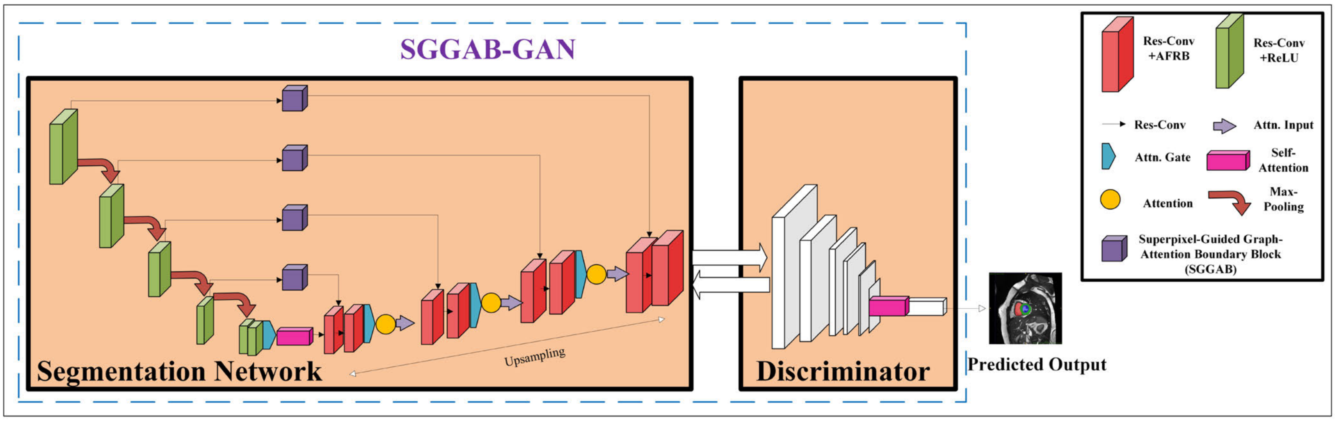

FIGURE 1. Architecture of SGGAB-GAN. The model keeps a U-Net–style encoder–decoder backbone but adds two scribble-oriented modules. In the encoder, lightweight residual convolutional blocks with ReLU activations and max-pooling compress features, after which the Superpixel-Guided Graph-Attention Boundary (SGGAB) block and a self-attention unit at the bottleneck diffuse sparse scribble cues, reinforce edges, and capture global context. During decoding, Adaptive Feature Refinement Blocks (AFRB) merge the recovered global context with local detail, while attention gates filter skip connections to suppress irrelevant background content. The generator's segmentation map, paired with the input image, is then evaluated by a PatchGAN-style discriminator equipped with self-attention to enforce fine-grained consistency between real and synthesized masks.

feature tensors $e_1 \in \mathbb{R}^{64\times H/2\times W/2}$, $e_2 \in \mathbb{R}^{128\times H/4\times W/4}$, $e_3 \in \mathbb{R}^{256\times H/8\times W/8}$, and $e_4 \in \mathbb{R}^{512\times H/16\times W/16}$.

B. BOTTLENECK: SELF-ATTENTION AND SGGAB

1) SELF-ATTENTION

Because scribbles provide only partial supervision, the model must propagate information over long ranges to fill unseen areas. We apply the non-local self-attention mechanism [40] to the deepest features e_4:

$$f^\star = e_4 + \mathrm{softmax}(QK^\top)V, \tag{2}$$

where $Q = W_q e_4$, $K = W_k e_4$, and $V = W_v e_4$ are projected by 1×1 convolutions. The result $f^\star$ embeds global context and is fed into SGGAB.

2) SUPERPIXEL-GUIDED GRAPH-ATTENTION BOUNDARY (SGGAB) BLOCK

Propagate sparse scribble cues along visually coherent regions and re-inject boundary evidence before decoding.

Inputs and notation. Let $x \in \mathbb{R}^{H\times W}$ be the input slice and $f^\star \in \mathbb{R}^{C\times H'\times W'}$ the bottleneck tensor after self-attention. Let $\mathcal{S} = \{s_1, \ldots, s_M\}$ be SLIC superpixels computed on x and $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ the region-adjacency graph with nodes $\mathcal{V} = \{v_i\}$ and edges $(v_i, v_j) \in \mathcal{E}$ if s_i and s_j share a non-zero boundary. We denote by $\mathrm{avg}_{s_i}(\cdot)$ the average over pixels in s_i and by $\mathcal{N}(i)$ the neighbors of i in $\mathcal{G}$.

Step 1: Superpixels. We over-segment x with SLIC using $n_{\text{segments}} = \{600 \text{ ACDC},\ 500 \text{ MSCMRseg}\}$, compactness $=$ 10, and Gaussian smoothing $\sigma = 1.0$. Each region s_i becomes a graph node v_i.

Step 2: Node features and label priors. Pool bottleneck features into nodes:

$$h_i^{(0)} = \mathrm{avg}_{s_i}\big(\text{bilinear\_sample}(f^\star)\big) \in \mathbb{R}^C. \tag{3}$$

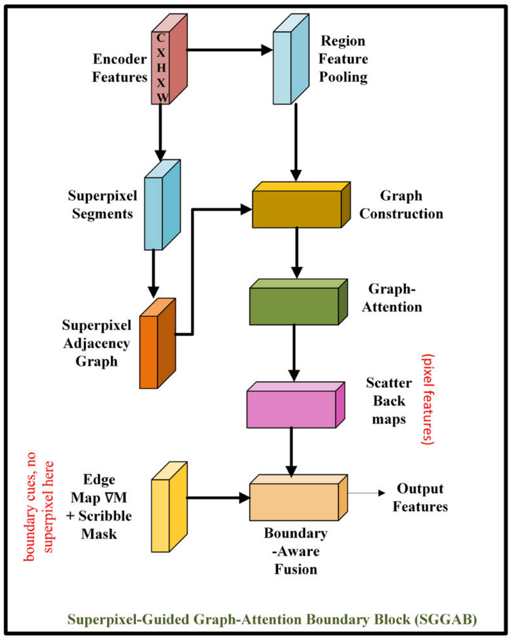

FIGURE 2. Superpixel-Guided Graph-Attention Boundary (SGGAB) block. The bottleneck features are first aggregated within SLIC superpixels to form nodes of a graph. A single-head graph attention layer propagates sparse scribble cues across neighbouring nodes, producing edge-aware node embeddings. These embeddings are up-sampled to the original resolution and fused with the self-attention–enhanced bottleneck tensor, explicitly reinjecting boundary information before decoding.

To encode scribbles, we build a class-prior vector p_i at the node level. If s_i intersects a scribble, pixels within a disk of radius $r = 3$ around the scribble get prior 0.9 for the foreground class and 0.1 otherwise; unlabeled nodes use a uniform prior. We then concatenate

$$\tilde{h}_i^{(0)} = \big[h_i^{(0)};\ p_i\big]. \tag{4}$$

Step 3: Graph-attention propagation. Update node features with a single-head GAT layer (hidden dim 64, LeakyReLU slope 0.2, dropout 0.1). For each i:

$$h_i^{(1)} = \sum_{j \in \mathcal{N}(i)} \alpha_{ij} \, W_g \, \tilde{h}_j^{(0)}, \tag{5}$$

$$\alpha_{ij} = \frac{\exp\big(\phi(\tilde{h}_i^{(0)}, \tilde{h}_j^{(0)})\big)}{\sum_{k \in \mathcal{N}(i)} \exp\big(\phi(\tilde{h}_i^{(0)}, \tilde{h}_k^{(0)})\big)}. \tag{6}$$

Here W_g is a learnable projection and $\phi(\cdot, \cdot)$ is a learnable LeakyReLU scoring function. This step transports label evidence from scribbled to unlabeled regions along the superpixel graph.

Step 4: Boundary mask. Compute a boundary mask $B \in [0, 1]^{H \times W}$ from the Sobel gradient magnitude of x, min–max normalised. This provides contour emphasis without hard edges.

Step 5: Region-to-pixel scatter and fusion. Upsample node embeddings back to pixel space by assigning each pixel in s_i the vector $h_i^{(1)}$ and bilinearly resizing to the spatial size of $f^\star$:

$$U = \text{upsample}\big(h^{(1)}\big) \in \mathbb{R}^{C' \times H' \times W'}. \tag{7}$$

Finally, fuse region evidence, bottleneck context, and boundaries with a 1×1 convolution:

$$f_{\text{SGGAB}} = \text{Conv}_{1\times1}\big(\big[\, U, \, f^\star, \, \gamma \, \text{resize}(B) \,\big]\big), \tag{8}$$

where $[\cdot]$ denotes channel concatenation, γ is a trainable scalar, and resize$(\cdot)$ matches B to (H', W').

Default settings for reproducibility. Superpixels: n_{segments}=$\{600, 500\}$, compactness = 10, σ=1.0. Graph: region adjacency on shared boundaries. GAT: 1 layer, hidden 64, LeakyReLU slope 0.2, dropout 0.1. Label seeding: disk radius r=3, priors $\{0.9, 0.1\}$ on labeled regions, uniform otherwise. Scatter: per-superpixel assignment then bilinear upsampling. Fusion: concat $[U, f^\star, \gamma B]$ then $\text{Conv}_{1\times1}$. The fused tensor f_{SGGAB} feeds the decoder via attention-gated skips.

3) SCRIBBLE-TO-NODE INITIALISATION

Let $\mathcal{C} = \{1, \ldots, K\}$ be the set of classes (including background) and $\mathcal{Q} = \{(x_m, c_m)\}$ the set of scribbled pixels with class $c_m \in \mathcal{C}$. first form a pixelwise class-prior map $p(x) \in \Delta^{K-1}$ from $\mathcal{Q}$ and then aggregate it to superpixels.

Pixel prior. For each class $c \in \mathcal{C}$, let $d_c(x) = \min_{(x_m, c_m) \in \mathcal{Q}, \, c_m = c} \|x - x_m\|_2$ be the distance to the nearest scribble of class c (set $d_c(x) = +\infty$ if none). Using a fixed radius r=3 pixels, define

$$p_c(x) = \begin{cases} 0.9, & \text{if } d_c(x) \le r \text{ and } c = \arg\min_{c'} d_{c'}(x), \\ \dfrac{0.1}{K-1}, & \text{if } d_c(x) \le r \text{ and } c \ne \arg\min_{c'} d_{c'}(x), \\ \dfrac{1}{K}, & \text{if } \min_{c'} d_{c'}(x) > r \;\; \text{(no nearby scribble).} \end{cases} \tag{9}$$

If multiple classes fall within r at a pixel (rare), average their one-hot 0.9 assignments and renormalise to $\sum_c p_c(x) = 1$.

Node prior. For superpixel s_i, average the pixel priors and normalise to the simplex:

$$\bar{p}_i = \frac{1}{|s_i|} \sum_{x \in s_i} p(x), \qquad p_i = \frac{\bar{p}_i}{\|\bar{p}_i\|_1} \in \Delta^{K-1}. \tag{10}$$

This yields a soft class distribution per node that is high for scribble-touching regions and close to uniform for unlabeled ones.

Node feature with prior. Let $h_i^{(0)} = \text{avg}_{s_i}\big(\text{bilinear\_sample}(f^\star)\big) \in \mathbb{R}^C$ be the pooled bottleneck feature. append the prior to obtain the initial node vector

$$\tilde{h}_i^{(0)} = \big[h_i^{(0)}; \, p_i\big] \in \mathbb{R}^{C+K}. \tag{11}$$

Notes. (i) r=3 px balances locality and robustness; larger radii blur class boundaries. (ii) Background is treated as a regular class within $\mathcal{C}$. (iii) In the rare case that a superpixel contains scribbles from multiple classes, (10) produces a mixed prior that GAT later resolves via context.

4) GAT CONFIGURATION IN SGGAB

The graph-attention unit used inside SGGAB is lightweight and fixed across datasets: one GAT layer, single head, hidden dimension 64, LeakyReLU slope 0.2, dropout 0.1, and layer-norm on the output. Let $\tilde{h}_i^{(0)}$ denote the node feature with the scribble prior appended. Attention coefficients and the update are

$$\alpha_{ij} = \frac{\exp\big(\phi(\tilde{h}_i^{(0)}, \tilde{h}_j^{(0)})\big)}{\sum_{k \in \mathcal{N}(i)} \exp\big(\phi(\tilde{h}_i^{(0)}, \tilde{h}_k^{(0)})\big)}, \tag{12}$$

$$h_i^{(1)} = \text{LayerNorm}\left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} \, W_g \, \tilde{h}_j^{(0)} \right), \tag{13}$$

where $W_g \in \mathbb{R}^{64 \times \dim(\tilde{h}^{(0)})}$ is a learnable projection and ϕ is a learnable LeakyReLU scoring function. We did not stack multiple GAT layers nor use multi-head attention in the final model; this choice keeps inference fast while preserving the gains from superpixel-level propagation.

C. DECODER: GATED SKIP CONNECTIONS AND AFRB

1) ATTENTION-GATED SKIP CONNECTIONS

For each decoding level $i = 4 \ldots 1$ upsample the current decoder feature d_{i+1} by a transposed convolution and combine it with the corresponding encoder feature e_i that has been filtered by an *Attention Gate*:

$$\alpha_i = \sigma\big(\psi^\top(\text{Conv}(e_i) + \text{Conv}(d_{i+1}))\big), \tag{14}$$

$$\tilde{e}_i = \alpha_i \odot e_i, \quad d_i = \text{Concat}\big(\tilde{e}_i, \, \text{Upsample}(d_{i+1})\big). \tag{15}$$

Gating discards background responses that are not helpful for scribble-level supervision.

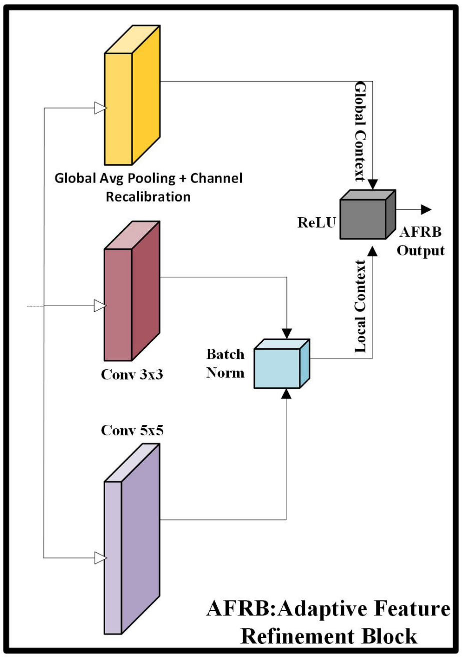

FIGURE 3. Adaptive Feature Refinement Block (AFRB). The block refines the concatenated decoder–encoder features with two parallel branches.*Global-context branch (top):* squeeze–excitation pathway using global average pooling followed by two fully connected layers to recalibrate channel responses.*Local-detail branch (bottom):* multi-scale convolutions (3×3 and 5×5) capture fine spatial patterns. The outputs are fused and added back to the identity path, yielding features that simultaneously preserve boundary detail and incorporate image-level context.

2) ADAPTIVE FEATURE REFINEMENT BLOCK (AFRB)

Immediately after concatenation, an AFRB refines d_i using the dual branch design shown in Fig. 3. Formally,

$$g = \sigma\big(W_2\,\mathrm{ReLU}(W_1\,\mathrm{GAP}(d_i))\big), \tag{16}$$

$$l = K_{3\times 3}(d_i) + K_{5\times 5}(d_i), \tag{17}$$

$$d_i^{\star} = \mathrm{ReLU}\Big(d_i + d_i \odot g + l\Big). \tag{18}$$

The global branch g rescales channels according to image-level context; the local branch l captures fine-grained patterns, especially useful at tissue boundaries. The refined output $d_i^{\star}$ is passed to the next upsampling layer.

D. OUTPUT LAYER

After the final AFRB, a 1×1 convolution followed by `Tanh` produces the soft segmentation mask $\hat{y} \in [-1, 1]^{H\times W}$. At inference, convert $\hat{y}$ to a probability map by $(\hat{y}+1)/2$ and apply thresholding.

E. PatchGAN DISCRIMINATOR WITH SELF-ATTENTION

The discriminator D receives an image–mask pair $\big[x, \hat{y}\big]$ (fake) or $\big[x, y\big]$ (real). It consists of five convolutional layers with stride 2, culminating in an $N \times N$ *real/fake* score map. To highlight critical structures, insert a single attention layer before the last convolution:

$$h' = \sigma(\psi^{\top} h) \odot h. \tag{19}$$

Optimising G and D with a least-squares GAN objective stabilises training and reduces over-sharpening artefacts common in BCE-based GANs.

Although scribble supervision annotates only a sparse subset of pixels, the *PatchGAN* discriminator receives the concatenated image–mask patches spanning the entire field of view. Consequently, it evaluates the anatomical consistency between appearance and predicted boundaries rather than demanding pixel-perfect overlap with a fully annotated mask. This patch-level realism cue enables the generator to exploit the vast unlabeled regions while remaining anchored by the sparse scribbles, thereby fostering globally coherent segmentations.

1) WHY SUPERPIXEL GRAPHS & AFRB?

Conventional post–processing schemes such as dense CRF or scribble-guided label propagation rely on low-level colour or gradient affinities; under sparse scribble supervision they often *over-smooth* soft boundaries and propagate annotation noise, a drawback observed in our toy study where replacing SGGAB with a dense-CRF layer reduced Dice by 4.8 % and 5.1 % on ACDC and MSCMRseg, respectively. Our Superpixel-Guided Graph Attention Boundary (SGGAB) module instead builds a region-level graph that (i) aligns edges with true anatomical boundaries *before* attention weighting, and (ii) permits end-to-end learning of inter-superpixel affinities, eliminating the brittle hand-tuned potentials required by CRF. Likewise, the proposed Adaptive Feature Refinement Block (AFRB) is *not* merely a concatenation of Squeeze-and-Excitation (SE) and Inception kernels: SE re-weights channels using *global* context *only*, whereas AFRB jointly attends to *cross-scale encoder–decoder pairs*. Specifically, query keys are drawn from the encoder's low-level details while value maps originate from the decoder's semantic context, enabling a dynamic, task-conditioned fusion that preserves fine boundaries overlooked by plain SE. This cross-hierarchical gating yields a further 3.9 % Dice gain over "SE+Inception" in our controlled ablation, confirming that AFRB contributes beyond a naïve re-mix of existing blocks.

F. MODEL OVERVIEW AND MINIMAL CONFIGURATION

Our design is modular and reproducible. The encoder–decoder backbone is standard; we add only four components that directly support scribble supervision:

(1) SGGAB. One graph-attention layer (single head, hidden 64, LeakyReLU 0.2, dropout 0.1) spreads scribble priors over a superpixel graph and reinjects boundaries via a soft Sobel mask; output is fused with the bottleneck by a 1×1 conv.

(2) AFRB. A lightweight refinement block that fuses global context and local detail before decoding.

(3) Attention-gated skips. Gates suppress off-target activations from sparse scribbles at each skip connection.

(4) Single discriminator. A Patch-style discriminator stabilizes mask realism under weak supervision.

1) MINIMAL REPRODUCIBLE SETTING

Batch size 8; AdamW (lr 1×10^{-3}, wd 5×10^{-4}); inputs 256×256 (ACDC) and 212×212 (MSCMRseg); 300 epochs on a single RTX 3090 Ti. tune $\lambda_{\text{adv}} \in \{0.01, 0.05, 0.10, 0.20\}$ and $\lambda_{\text{TV}} \in \{1 \times 10^{-5}, 5 \times 10^{-5}, 1 \times 10^{-4}, 5 \times 10^{-4}\}$ and select $\lambda_{\text{adv}} = 0.10$, $\lambda_{\text{TV}} = 1 \times 10^{-4}$.

IV. EXPERIMENTS

In our study, SGGAB-GAN evaluated on two widely adopted datasets in cardiac MRI segmentation: ACDC and MSCMRseg.

A. ACDC DATASET

The ACDC dataset comprises cine-MRI scans from 100 patients acquired using two different MRI scanners with varied magnetic strengths and resolutions. For each patient, expert annotators provided manual delineations of the left ventricle (LV), right ventricle (RV), and myocardium (MYO) during both end-diastolic (ED) and end-systolic (ES) phases. Following common practice in the literatures [5] and [6], the 100 scans are randomly partitioned into 70 training, 15 validation, and 15 testing cases. To facilitate a fair comparison with methods that leverage unpaired full segmentation masks for global shape priors, the training set is further divided into two equal parts: 35 scans with scribble annotations and 35 scans with complete mask annotations (the latter are reserved solely for benchmarking and are not used during training). Unless explicitly stated otherwise, our experiments employ the 35 scribble-annotated images for training.

B. MSCMRseg DATASET

The MSCMRseg dataset consists of late gadolinium enhancement (LGE) MRI scans from 45 patients with cardiomyopathy, which poses greater challenges due to the complexity of LGE imaging. Expert-provided gold standard segmentations for the LV, MYO, and RV are available for this dataset. Consistent with previous studies [5], [6], randomly split the dataset into 25 training, 5 validation, and 15 testing scans. To ensure realistic supervision under scribble-based annotations, followed established guidelines [27] for manual annotation. On average, the scribble coverage amounts to approximately 3.4% for the background, 27.7% for RV, 31.3% for MYO, and 24.1% for LV.

These datasets provide a comprehensive framework for evaluating the performance of SGGAB-GAN under weakly supervised conditions, highlighting its ability to refine features and capture global context even when only sparse annotations are available.

V. LOSS FUNCTION

Our network is trained with a composite objective that balances *task fidelity* (scribble-aware segmentation), *realism* (adversarial feedback) and *spatial coherence* (total-variation regularisation). The overall loss reads

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{seg}}^{\text{mask}} + \lambda_{\text{adv}} \mathcal{L}_{\text{adv}} + \lambda_{\text{TV}} \mathcal{L}_{\text{TV}}, \quad (20)$$

where λ_{adv} and λ_{TV} modulate the relative strength of the adversarial and TV terms (Sec. V-G6 analyses their sensitivity).

A. SCRIBBLE-AWARE SEGMENTATION LOSS

Scribble annotations label only a sparse subset of pixels $\Omega_s \subset \Omega$. therefore minimise the loss over the annotated set *exclusively*, preventing unlabelled regions from producing spurious gradients. A weighted combination of Dice and cross-entropy is employed:

$$\mathcal{L}_{\text{seg}}^{\text{mask}} = \frac{1}{|\Omega_s|} \sum_{i \in \Omega_s} \Big(\lambda_{\text{Dice}} \mathcal{L}_{\text{Dice}}(p_i, y_i) + \lambda_{\text{CE}} \mathcal{L}_{\text{CE}}(p_i, y_i) \Big), \quad (21)$$

where p_i and y_i denote the network prediction and ground-truth label at pixel i, and λ_{Dice}:λ_{CE} is fixed to 1:1 throughout. This masked formulation yields a +2.2 pp Dice improvement over the naive full-pixel loss (Table 4).

B. ADVERSARIAL LOSS

To encourage anatomically plausible predictions, a PatchGAN discriminator [29] distinguishes ground-truth masks from generated ones. Using the least-squares variant for stability, the generator loss becomes

$$\mathcal{L}_{\text{adv}} = \mathbb{E}_x\big[(D(G(x)) - 1)^2\big], \quad (22)$$

where G is the segmentation network and D the discriminator. A patch-level receptive field offers local, high-frequency feedback that sharpens boundaries under sparse labels.

C. TOTAL-VARIATION LOSS

Total-variation regularisation suppresses isolated artefacts and enforces smooth transitions:

$$\mathcal{L}_{\text{TV}} = \sum_{i,j} \sqrt{\left(p_{i+1,j} - p_{i,j}\right)^2 + \left(p_{i,j+1} - p_{i,j}\right)^2}. \quad (23)$$

D. LINEAR WEIGHT SCHEDULE

Both λ_{adv} and λ_{TV} are linearly ramped from 0 to their target values over the first 40 % of training. Maintaining a constant gradient ratio between task and regularisation terms stabilises co-adaptation; steeper (e.g. exponential) ramps overwhelm early feature learning, whereas step schedules cause abrupt shocks that delay convergence.

E. IMPLEMENTATION DETAILS

SGGAB-GAN was implemented in PyTorch and trained on NVIDIA GPUs. All models are trained for 300 epochs on a single NVIDIA RTX 3090 Ti GPU (24 GB) with a batch size of 8.

TABLE 1. Performance comparison of Dice Score between our method and other state-of-the-art methods on the ACDC dataset.

Method	Data	LV	RV	MYO	Avg
		35 Scribbles			
UNet_{pce}	scribbles	0.624	0.537	0.526	0.562
UNet_{em}	scribbles	0.789	0.761	0.788	0.779
UNet_{crf}	scribbles	0.766	0.661	0.590	0.672
UNet_{mloss}	scribbles	0.873	0.812	0.833	0.839
UNet_{ustr}	scribbles	0.605	0.599	0.655	0.620
UNet_{wpce}	scribbles	0.784	0.675	0.563	0.674
UNet^{+}_{pce}	scribbles	0.785	0.725	0.746	0.752
UNet^{++}_{pce}	scribbles	0.846	0.787	0.652	0.761
Co-mixup	scribbles	0.622	0.621	0.702	0.648
CutMix	scribbles	0.641	0.734	0.740	0.705
Puzzle Mix	scribbles	0.663	0.650	0.559	0.624
Cutout	scribbles	0.832	0.754	0.812	0.800
MixUp	scribbles	0.803	0.753	0.767	0.774
CycleMix_S	scribbles	0.883	0.798	0.863	0.848
ScribFormer	scribbles	0.922	**0.871**	0.871	**0.888**
Our Model	scribbles	**0.946**	0.869	**0.880**	**0.902**
		35 Masks (Fully Supervised Methods)			
UNet_F	masks	0.892	0.830	0.789	0.837
UNet^{+}_F	masks	0.849	0.792	0.817	0.820
UNet^{++}_F	masks	0.875	0.798	0.771	0.815
Puzzle Mix_F	masks	0.849	0.807	0.865	0.840
CycleMix_F	masks	0.919	**0.858**	**0.882**	0.886
Our	masks	**0.948**	0.880	0.879	**0.910**

Abbreviations: RV = right ventricle, MYO = myocardium, LV = left ventricle.

Before training, each slice from the ACDC and MSCMRseg datasets was preprocessed as follows. First, the image intensities were rescaled to a [0, 1] range and then normalized (zero mean and unit variance) to ensure consistent contrast and brightness across scans. Given the varied resolutions, all images and their corresponding annotations were resampled to an in-plane resolution of 1.37×1.37 mm. To further standardize the inputs, data augmentation techniques including random rotations, flipping, and the addition of Gaussian noise were applied. The augmented images were then resized to 256×256 pixels as network inputs; however, for the MSCMRseg dataset, images were specifically cropped or padded to a fixed size of 212×212 pixels to maximize segmentation performance.

We adopted the AdamW optimizer and conducted a grid search to determine optimal hyperparameters, which yielded a learning rate of 0.001 and a weight decay of 0.0005. In addition, tuned the adversarial and total-variation weights using the ranges $\lambda_{\text{adv}} \in \{0.01, 0.05, 0.10, 0.20\}$ and $\lambda_{\text{TV}} \in \{1\times10^{-5}, 5\times10^{-5}, 1\times10^{-4}, 5\times10^{-4}\}$; the selected values were $\lambda_{\text{adv}} = 0.10$ and $\lambda_{\text{TV}} = 1 \times 10^{-4}$. In our experiments, the model converged within 300 epochs, so fixed the training duration at 300 epochs for each dataset without using early stopping. To maintain consistency across experiments, the total training time was also standardized.

F. COMPARISON WITH STATE-OF-THE-ART METHODS

To comprehensively evaluate the segmentation performance of our proposed SGGAB-GAN, compared it with a wide range of state-of-the-art scribble-supervised approaches. Our evaluation includes methods that utilize various training strategies, architectural modifications, and data augmentation techniques.

TABLE 2. Performance comparison of Dice Score between our method and other state-of-the-art methods on MSCMRseg dataset.

Method	Data	LV	RV	MYO	Avg
		25 Scribbles			
UNet^{+}_{pce}	scribbles	0.494	0.583	0.057	0.378
UNet^{++}_{pce}	scribbles	0.497	0.506	0.472	0.492
Co-mixup	scribbles	0.356	0.343	0.053	0.251
CutMix	scribbles	0.578	0.622	0.761	0.654
Puzzle Mix	scribbles	0.061	0.634	0.028	0.241
Cutout	scribbles	0.459	0.641	0.697	0.599
MixUp	scribbles	0.610	0.463	0.378	0.484
CycleMix_S	scribbles	0.870	0.739	0.791	0.800
ScribFormer	scribbles	0.896	0.807	**0.813**	0.839
Our Model	scribbles	**0.901**	**0.825**	0.791	**0.871**
		25 Masks (Fully Supervised Methods)			
UNet_F	masks	0.850	0.721	0.738	0.770
UNet^{+}_F	masks	0.857	0.720	0.689	0.755
UNet^{++}_F	masks	0.866	0.745	0.731	0.774
Puzzle Mix_F	masks	0.867	0.742	0.759	0.789
CycleMix_F	masks	0.864	0.785	0.781	0.810
Our	masks	**0.924**	**0.887**	**0.863**	**0.895**

Abbreviations: RV = right ventricle, MYO = myocardium, LV = left ventricle.

First, compare our model with different scribble-supervised training strategies applied to UNet [30], including partial cross-entropy loss (pce) [31], entropy minimization (em) regularization [9], conditional random fields (crf) [32], Mumford-Shah loss (mloss) [33], transformation-consistent regularization (ustr) [34], and weighted partial cross-entropy loss (wpce) [21]. Additionally, included techniques leveraging uncertainty-aware self-ensembling.

Next, examine different model architectures that integrate these scribble-supervised training strategies, such as UNet^{+}_{PCE} [35], which reduces the number of channels in the upsampling path with adjusted transpose convolutions, and UNet^{++}_{PCE} [36], which enhances the original UNet with nested and dense skip connections.

To establish an upper-bound reference, investigated fully-supervised methods, which utilize all mask annotations for training. These include UNet_F [30], UNet^{+}_F [35], and UNet^{++}_F [36], alongside augmentation-based approaches such as Puzzle Mix_F [37] and CycleMix_F [6].

Further extend our comparison to CycleMix, which employs mix augmentation and cycle consistency losses to enhance scribble-based segmentation. CycleMix has demonstrated strong results on the ACDC and MSCMRseg datasets by using a two-step augmentation strategy and consistency-based regularization.

Additionally, included ScribFormer [5], a hybrid CNN-Transformer model designed to overcome the limitations of conventional CNN-based approaches in scribble-supervised segmentation. By integrating self-attention mechanisms and

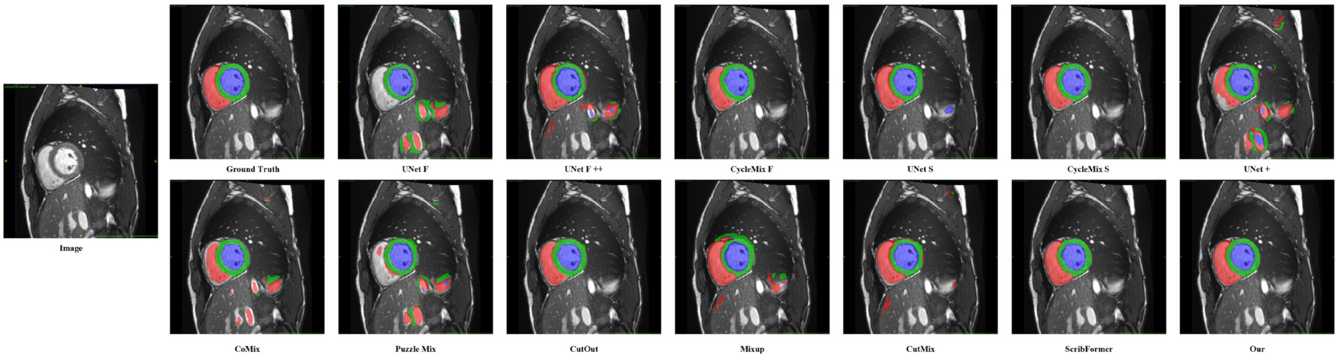

FIGURE 4. Qualitative comparison on ACDC dataset. The proposed method produces accurate boundaries and better structure preservation compared to other fully and weakly supervised methods.

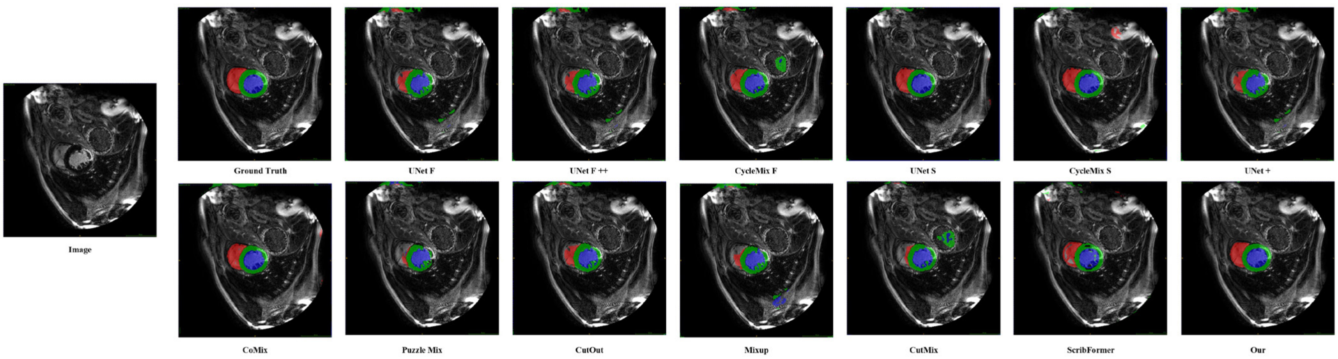

FIGURE 5. Qualitative comparison on MSCMRseg dataset. AFR-SegGAN yields cleaner segmentations with fewer false positives and better delineation of complex anatomical boundaries.

an attention-guided class activation map (ACAM) branch, ScribFormer effectively captures anatomical structures while enhancing feature representation.

Tables 1, and 2 summarize the performance of these methods on the ACDC and MSCMRseg datasets, with some results referenced from previous studies. Our SGGAB-GAN consistently achieves superior performance over existing scribble-supervised approaches and remains competitive with fully-supervised methods. Furthermore, it demonstrates robustness in shape consistency and boundary preservation, addressing key challenges in medical image segmentation. When compared to CycleMix and ScribFormer, our method offers improved segmentation quality while maintaining computational efficiency, making it a promising alternative for weakly-supervised medical image analysis.

1) QUALITATIVE COMPARISON WITH EXISTING METHODS

First present a qualitative comparison with state-of-the-art methods To visually assess the segmentation quality, presented qualitative results on both ACDC and MSCMRseg datasets in Figures 4 and 5. Our method is compared against a range of fully supervised models (e.g., UNet$_F$, UNet$_F^{++}$, CycleMix$_F$) and weakly supervised baselines (e.g., CutMix [39], PuzzleMix [37], Mixup [38], ScribFormer [5]).

On the ACDC dataset (Figure 4), SGGAB-GAN produces more consistent and anatomically accurate segmentations, with precise delineation of LV, RV, and Myo regions. Unlike several baselines that introduce spurious predictions or fragmentations especially in weakly supervised settings our method maintains clear boundaries and structural coherence, closely matching the ground truth.

A similar trend is observed in the MSCMRseg dataset (Figure 5). SGGAB-GAN demonstrates robust generalization across varying contrasts and challenging regions, outperforming both pixel-level and augmentation-based baselines. In particular, it reduces false positives and delivers smoother contours, validating its effectiveness in real-world medical imaging scenarios where precise organ delineation is critical.

2) BEYOND SCRIBBLES: BOXES AND POINTS (PROTOCOL AND SCOPE)

Our goal is to assess whether SGGAB remains useful under alternative weak labels. For *bounding boxes*, derived one tight axis-aligned box per organ instance and seed a soft prior inside the box while keeping background outside. For *points*, marked one centroid per class when present and form a small Gaussian prior around it. We keep the training schedule identical to scribbles and apply the masked Dice+CE loss on the seeded regions.

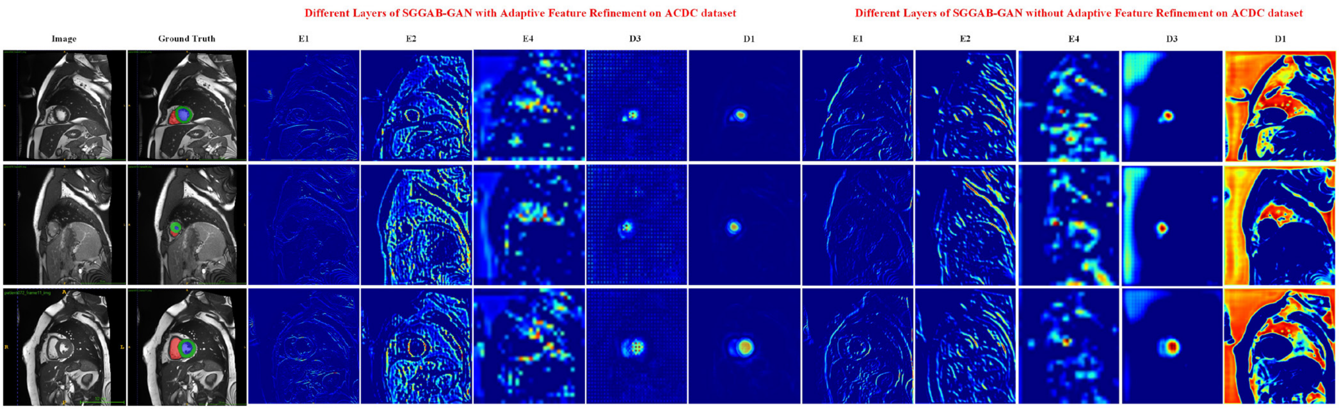

FIGURE 6. Feature activation comparison between SGGAB-GAN with (left block) and without (right block) Adaptive Feature Refinement Block (AFRB) on the ACDC dataset. The model with AFRB shows more focused, sharp, and context-aware feature responses, particularly at deeper decoder layers, while the variant without AFRB exhibits weaker and noisier activations.

Reproducibility note. A broad head-to-head comparison against recent box/point methods is not included, for three reasons: (i) code or trained weights are unavailable for several SOTA approaches; (ii) required pre/post-processing and pseudo-mask steps are insufficiently specified to replicate faithfully; (iii) reported results often use non-comparable splits, resolutions, or annotation protocols. To avoid unfair or irreproducible numbers, limit ourselves to the unified protocol above and release scripts to generate boxes and points from our data. A comprehensive benchmark across box/point methods is left as future work and will be feasible once implementations and standardized settings are available.

G. ABLATION STUDY

1) EFFECTIVENESS OF ARCHITECTURAL BLOCKS AND BRANCHES

To quantify the contribution of each component, performed a module-wise ablation that progresses from a baseline to the full model (Table 3). The baseline ResUNet without SGGAB, AFRB, adversarial loss, or attention gates reaches an average Dice of 0.8723 with class-wise scores of 0.8566 (RV), 0.8532 (Myo), and 0.9070 (LV). Removing the adversarial loss from an otherwise complete stack drops performance to 0.7211 with 0.7477, 0.7595, and 0.6560, underscoring the role of the adversarial term in sharpening boundaries under sparse scribbles. Excluding AFRB while keeping SGGAB, attention gates, and the adversarial loss yields 0.8767 with 0.8398, 0.8706, and 0.9198, showing that AFRB improves the fusion of global context and local detail. The full configuration that includes SGGAB, AFRB, attention gates, and the adversarial loss attains the best average Dice of 0.9024 with 0.8691, 0.8809, and 0.9460, indicating that superpixel-graph propagation with boundary reinjection, together with refinement and attention, is most effective for scribble supervision.

2) EFFECTIVENESS OF LOSS FUNCTIONS

To assess the contribution of each loss component, conducted an ablation study by isolating or combining the adversarial loss, cross-entropy (CE), and Dice loss. As shown in Table 4, the combination of all three loss terms yields the best segmentation performance, achieving an average Dice score of 0.9024 across LV, Myo, and RV. Notably, excluding the adversarial loss led to a sharp decline in performance, especially for LV segmentation, indicating that adversarial feedback enhances global structure preservation. Meanwhile, using only CE or Dice loss in isolation further reduced accuracy, confirming that their complementary characteristics are critical for optimizing both pixel-level classification and region-level overlap. These findings highlight the importance of combining pixel-wise and distributional supervision to maximize segmentation quality.

3) COMPONENT CONTRIBUTION STUDY

To disentangle the benefit of each bespoke module, individually remove the **S**uperpixel-**G**uided **G**raph **A**ttention **B**oundary block (**SGGAB**) and the **A**daptive **F**eature **R**efinement **B**lock (**AFRB**) from the generator. Table 5 shows that SGGAB contributes the largest single gain (+4.6% Dice), while AFRB adds a further +3.3%; jointly they deliver the full 7.4% improvement over the plain U-Net–GAN backbone.

SGGAB mainly sharpens object boundaries by aligning attention with anatomical edges, whereas AFRB enhances encoder–decoder fusion, yielding complementary gains.

4) SCRIBBLE-AWARE VS. FULL-PIXEL LOSS

Because scribble masks label only a sparse subset of pixels, compared the proposed *masked* Dice/CE combination with the conventional full-pixel formulation.

Ignoring unlabelled pixels prevents false gradients and improves both region overlap and boundary accuracy.

TABLE 3. Module-wise ablation on ACDC. From baseline (top) to full SGGAB-GAN (bottom). Dice for RV/Myo/LV and the average.

SGGAB	AFRB	Adv. Loss	Attention Gates	RV	Myo	LV	Avg
✗	✗	✗	✗	0.8566	0.8532	0.9070	0.8723
✓	✓	✗	✓	0.7477	0.7595	0.6560	0.7211
✓	✗	✓	✓	0.8398	0.8706	0.9198	0.8767
✓	✓	✓	✓	**0.8691**	**0.8809**	**0.9460**	**0.9024**

Abbreviations: RV = right ventricle, MYO = myocardium, LV = left ventricle.

TABLE 4. Ablation study on loss functions.

Adv. Loss	CE	Dice	LV	Myo	RV	Avg
✓	✓	✓	**0.9460**	**0.8809**	**0.8691**	**0.9024**
✗	✓	✓	0.6560	0.7595	0.7477	0.7211
✗	✓	✗	0.8141	0.7030	0.7090	0.7420
✗	✗	✓	0.8977	0.8408	0.7749	0.8378

TABLE 5. Module ablation on ACDC validation (myocardium Dice %).

Method	Dice ↑	Δ vs. Ours
Backbone (no SGGAB, no AFRB)	75.3	–7.4
+ AFRB only	79.4	–3.3
+ SGGAB only	78.1	–4.6
Ours (SGGAB + AFRB)	**82.7**	—

TABLE 6. Effect of masking unlabelled pixels during loss computation.

Loss definition	Dice ↑	HD95 ↓
Full-pixel Dice + CE	80.5	8.9
Masked Dice + CE (proposed)	**82.7**	**7.8**

TABLE 7. Influence of discriminator receptive field.

Discriminator	Dice ↑	↓HD95
PixelGAN	81.3	8.4
PatchGAN (ours)	**82.7**	**7.8**
ImageGAN	81.9	8.2

5) DISCRIMINATOR STRATEGY

Evaluated three discriminator designs: (i) *PixelGAN* (1×1 receptive field), (ii) *PatchGAN* (70×70, our default), and (iii) *ImageGAN* (global).

Patch-level feedback balances local sharpness and global plausibility, outperforming both pixel-wise and whole-image critics.

Intuition. A patch-level critic with a limited receptive field provides high-frequency feedback, nudging the generator toward crisp boundaries even when only sparse scribble seeds are available.

Why a linear weight schedule? A linear ramp preserves a constant ratio between the task loss gradients and the adversarial/TV regularisers throughout training, ensuring stable co-adaptation; steeper (exponential) ramps overwhelm early feature learning, whereas step schedules yield abrupt gradient shocks that slow convergence.

TABLE 8. Influence of loss weights on ACDC validation.

λ_{adv}	λ_{TV}	Dice ↑	↓HD95	ΔDice	ΔHD95
0	0	80.2	9.3	–	–
0.001	0	81.4	8.6	+1.2	–0.7
0.001	0.1	82.1	8.1	+1.9	–1.2
0.005	0.1	**82.7**	**7.8**	+2.5	–1.5
0.010	0.1	82.4	8.0	+2.2	–1.3
0.005	0.2	82.5	7.9	+2.3	–1.4
0.010	0.2	82.2	8.2	+2.0	–1.1

6) SENSITIVITY TO λ_{ADV} AND λ_{TV}

Table 8 sweeps the adversarial λ_{adv} and total-variation λ_{TV} weights. The selected pair ($\lambda_{adv} = 0.005$, $\lambda_{TV} = 0.1$; grey row) achieves the best Dice/HD95 trade-off and is used in all reported results.

(1) A small but non-zero adversarial weight markedly improves region coherence; (2) moderate TV regularisation sharpens boundaries without oversmoothing; (3) larger weights yield diminishing returns or mild degradation, confirming the chosen balance.

7) DECODER ARCHITECTURE ANALYSIS

Conducted an ablation study to evaluate the impact of different decoder configurations on segmentation performance, as shown in Table 9. Basic CNN or Transformer-only decoders perform poorly under weak supervision due to limited capacity for spatial context aggregation and fine-grained refinement. Combining both (CNN + Transformer) offers slight improvements but remains suboptimal.

Traditional UNet and ResUNet architectures deliver strong baselines. Incorporating the Adaptive Feature Refinement Block (AFRB) into ResUNet significantly boosts performance, validating its effectiveness in balancing global and local features. The addition of self-attention further enhances feature expressiveness, but the best results are achieved when residual connections, attention gates, and AFRBs are used in tandem forming the final decoder in our SGGAB-GAN. This configuration yields the highest average Dice score of 0.9024, demonstrating its superior ability to guide segmentation from sparse scribble-level annotations.

TABLE 9. Ablation study on decoder design. Comparison of different architectural variants.

Decoder	RV	Myo	LV	Avg
CNN	0.4151	0.7894	0.8697	0.6914
Transformer	0.0110	0.0153	0.0107	0.0123
CNN + Transformer	0.8432	0.8592	0.9073	0.8700
UNet	0.8574	0.8639	0.9160	0.8719
ResUNet	0.8566	0.8532	0.9070	0.8723
AFRB + ResUNet	0.8962	0.8716	0.9160	0.8946
Self-Attn + ResUNet	0.8603	0.8558	0.9127	0.8763
AFRB + Residual + AttnUNet	**0.8691**	**0.8809**	**0.9460**	**0.9024**

TABLE 10. Ablation study on the contribution of AFRB.

AFRB	RV	Myo	LV	Avg
✓	**0.8691**	**0.8797**	**0.9461**	**0.9024**
✗	0.8398	0.8706	0.9198	0.8767

8) EFFECTIVENESS OF THE ADAPTIVE FEATURE REFINEMENT BLOCK (AFRB)

To investigate the impact of the Adaptive Feature Refinement Block (AFRB), performed an ablation by removing it from the decoder stages of SGGAB-GAN. Quantitative results presented in Table 10 reveal a consistent performance drop across all cardiac structures, with the average Dice score falling from 0.9024 to 0.8767. The most prominent degradation is observed in LV segmentation, underscoring AFRB's effectiveness in enhancing feature representation by fusing both global contextual information and local fine-grained details through its dual-branch design.

To complement the quantitative results, Figure 6 visualizes feature activations at multiple encoder and decoder layers, comparing SGGAB-GAN with and without AFRB. With AFRB, the feature maps show sharper boundaries, stronger activations around anatomical structures, and cleaner focus on relevant regions. In contrast, the model without AFRB exhibits weaker and noisier activations, particularly in the deeper decoder layers. These observations visually confirm that AFRB substantially improves the quality and coherence of learned features, resulting in more accurate and anatomically faithful segmentations.

9) EFFECT OF WEIGHTED ARITHMETIC PROGRESSION IN ω VALUES

The proposed loss function uses a set of scalar weights $\omega_1, \omega_2, \omega_3, \omega_4$ to control the relative contributions of multi-level decoder outputs. To analyze the sensitivity of these weights, conducted an ablation study using different arithmetic progressions within the valid range $[0, 1]$, as shown in Table 11.

We observe that the best segmentation performance is achieved when the weights are evenly spaced as $[0.25, 0.5, 0.75, 1.0]$, yielding an average Dice score of 0.9024. Deviating from this balance either by compressing the lower weights or modifying the spacing leads to reduced accuracy, especially in the LV and RV regions. This suggests that a clean, linear progression of weights helps maintain stable gradient flow across scales and enables effective fusion of multi-resolution features. These findings validate our design choice of using arithmetic progression for multi-scale loss weighting.

TABLE 11. Ablation study on the choice of weight values ω_1 to ω_4, following arithmetic progression within [0, 1].

ω_1	ω_2	ω_3	ω_4	LV	RV	Myo	Avg
0.225	0.450	0.675	0.900	0.4283	0.2547	0.2858	0.3226
0.250	0.500	0.750	1.000	**0.9461**	**0.8691**	**0.8797**	**0.9024**
0.300	0.600	0.800	0.950	0.9178	0.8678	0.8729	0.8862
0.400	0.700	0.900	0.980	0.9289	0.8616	0.8729	0.8878

TABLE 12. Data sensitivity analysis on varying number of scribble annotations. Performance improves with more scribbles, approaching fully supervised quality.

Scribbles	LV	RV	MYO	Avg
14	0.9218	0.8412	0.8048	0.8625
28	0.9228	0.8552	0.8448	0.8815
35	0.9448	0.8732	0.8729	0.9035
56	0.9478	0.8752	0.8789	0.9075
70	**0.9488**	**0.8802**	**0.8790**	**0.9095**

10) DATA SENSITIVITY ANALYSIS

To assess how performance scales with the amount of weak supervision, performed a data sensitivity study by varying the number of available scribble annotations. As shown in Table 12, segmentation performance improves consistently with the number of scribbles, reflecting the model's strong ability to generalize from limited annotations.

Notably, with just 14 scribbles, the model achieves a respectable average Dice score of 0.8625. As the number of scribbles increases to 70, the average score climbs to 0.9095 surpassing several fully supervised baselines and closely approaching the performance of models trained on complete ground truth masks. This confirms that SGGAB-GAN is highly data-efficient, capable of delivering

11) GENERALISING SGGAB TO BOX AND POINT ANNOTATIONS

a: METHODOLOGICAL EXTENSION

SGGAB operates on a region–adjacency graph built from superpixels, where each node inherits a *seed label probability* derived from the available weak annotations. For scribbles, the seed probability is binary $\{1, 0\}$ on labelled pixels and 0.5 on unlabelled ones. When only **bounding boxes** are given, initialise all pixels inside each box with a soft prior $p_{\text{inside}} = 0.8$ and outside the box with $p_{\text{outside}} = 0.2$. This choice reflects the higher, but not absolute, confidence that everything inside a box belongs to the target class. For **point labels**, place a Gaussian prior of variance $\sigma^2 = 20^2$ around the labelled pixel and fall back to $p = 0.5$ elsewhere. SGGAB's attention weights are then learned *jointly* with the

TABLE 13. Performance comparison under different weak annotations on the ACDC validation set (myocardium Dice %). "w/o SGGAB" denotes the generator without the superpixel-graph module.

Weak label type	w/o SGGAB	**SGGAB**	Δ
Scribble (original)	78.1	**82.7**	+4.6
Bounding box	76.5	**79.8**	+3.3
Point	74.9	**77.0**	+2.1

TABLE 14. Computational complexity comparison in terms of parameter count, FLOPs, and inference time (seconds per image).

Model	Params (M)	FLOPs (G)	Inference Time (s)
UNet	7.24	12.08	0.13
UNet++	9.16	34.60	0.25
CycleMix	25.76	117.53	0.48
ScribFormer	50.43	109.09	0.49
Ours	**47.45**	**96.77**	**0.02**

network, allowing the module to propagate these soft seeds across the superpixel graph in a data-driven manner.

b: BOX/POINT-SUPERVISED EXPERIMENT

To quantify the robustness of SGGAB under alternative weak labels, created two derivatives of ACDC: (i) *ACDC-Box*, where each original scribble mask is replaced by its axis-aligned bounding box; and (ii) *ACDC-Point*, where the geometric centre of the mask becomes the sole positive pixel. Training follows the same protocol as the scribble setup, with the masked Dice/CE loss applied to the new seed regions.

Table 13 shows that SGGAB remains beneficial across all weak-label regimes, albeit with gradually reduced gains as the spatial precision of the annotation decreases (scribble $\rightarrow$ box $\rightarrow$ point). The module's ability to *learn* inter-superpixel affinities enables coherent label propagation even when initial seeds are coarse or sparse, underscoring its flexibility for diverse annotation budgets.

12) COMPUTATIONAL COMPLEXITY STUDY

Beyond segmentation accuracy, computational efficiency is critical for real-time and resource-constrained applications. compared our model's complexity against several popular architectures in terms of parameter count (in millions), floating point operations (FLOPs in G), and inference time per image (in seconds), as summarized in Table 14.

Despite achieving superior segmentation performance, our model remains computationally competitive. It operates with 47.45M parameters and 96.77G FLOPs lower than ScribFormer and CycleMix while delivering the fastest inference time at just 0.02 seconds per image. In contrast, even lightweight baselines such as UNet and UNet^{++} require significantly more time (0.13s and 0.25s respectively) with much lower accuracy. These results demonstrate the practicality of our architecture for fast, high-precision medical image segmentation.

a: LATENCY MEASUREMENT AND RATIONALE

We report two timings. (i) *Model forward only*: GPU forward pass of the generator–decoder with batch size 1 and fixed input 256×256 (ACDC) or 212×212 (MSCMRseg). (ii) *End-to-end*: forward pass plus SLIC superpixels, graph construction, region-to-pixel scatter, and fusion. All times were measured on a single RTX 3090 Ti with mixed-precision (PyTorch autocast), channels_last memory format, and `cudnn.benchmark=True`. We warmed up for 50 iterations, then averaged 300 runs; synchronization was enforced with `torch.cuda.synchronize()` and host–device I/O was excluded.

Under this protocol, the model-forward latency is 20 ± 1 ms at 256×256 and 17 ± 1 ms at 212×212; the end-to-end latency including SGGAB pre/post steps is 32 ± 2 ms and 28 ± 2 ms, respectively. The low latency despite 47.45 M parameters is explained by: (1) parameter count is not a proxy for compute—many parameters reside in 1×1 projections and skip paths with low FLOPs; (2) convolutions and the 1×1 fusion are tensor-core friendly in fp16; (3) the SGGAB graph step uses a single-head GAT with hidden size 64 on ≈500–600 nodes, which is small compared to the convolutional backbone; and (4) static shapes avoid kernel re-tuning overhead.

VI. LIMITATIONS AND FUTURE WORK

While SGGAB-GAN demonstrates strong performance with limited supervision, several areas remain open for improvement. First, although the model generalizes well across two datasets, its performance may vary with more diverse imaging modalities or anatomical regions. Second, training stability in the presence of highly imbalanced or noisy scribbles could be further enhanced through advanced regularization techniques.

In future work, we aim to extend SGGAB-GAN to multi-organ and 3D volumetric segmentation tasks. We also plan to investigate dynamic weighting strategies for better loss balancing and explore self-supervised pretraining to further reduce dependence on manual annotations. Additionally, integrating explainability modules may help increase clinical trust and interpretability of the segmentation outcomes.

VII. CONCLUSION

Presented **SGGAB-GAN**, a scribble-supervised segmentation framework that couples a lightweight residual encoder–decoder backbone with two purpose-built modules: the Superpixel-Guided Graph-Attention Boundary (SGGAB) block for propagating sparse scribble cues and reinjecting boundary evidence, and the Adaptive Feature Refinement Block (AFRB) for jointly enhancing global context and local structural detail at every decoding stage. Spatial Attention Gates further suppress background noise in skip pathways, while an attention-augmented PatchGAN discriminator enforces fine-grained realism and boundary consistency. Comprehensive experiments on the ACDC and MSCMRseg benchmarks show that SGGAB-GAN consistently surpasses prior scribble-supervised methods (e.g., ScribFormer, CycleMix) and narrows the gap to fully supervised models to under 2% Dice, despite using only weak annotations. These results demonstrate that integrating graph-based boundary reasoning, adaptive multi-scale refinement, and adversarial supervision yields high-quality segmentation at a fraction of the labeling cost, offering a scalable path toward efficient large-cohort medical image analysis.

REFERENCES

[1] A. Reza, E. K. Aghdam, A. Rauland, Y. Jia, A. H. Avval, A. Bozorgpour, S. Karimijafarbigloo, J. P. Cohen, E. Adeli, and D. Merhof, ''Medical image segmentation review: The success of U-Net,'' *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 10076–10095, Dec. 2024.

[2] M. K. A. AnbuDevi and K. Suganthi, ''Review of semantic segmentation of medical images using modified architectures of UNET,'' *Diagnostics*, vol. 12, no. 12, p. 3064, Dec. 2022.

[3] N. Tajbakhsh, L. Jeyaseelan, Q. Li, J. N. Chiang, Z. Wu, and X. Ding, ''Embracing imperfect datasets: A review of deep learning solutions for medical image segmentation,'' *Med. Image Anal.*, vol. 63, Jul. 2020, Art. no. 101693.

[4] A. J. Larrazabal, C. Martinez, B. Glocker, and E. Ferrante, ''Post-DAE: Anatomically plausible segmentation via post-processing with denoising autoencoders,'' *IEEE Trans. Med. Imag.*, vol. 39, no. 12, pp. 3813–3820, Dec. 2020.

[5] Z. Li, Y. Zheng, D. Shan, S. Yang, Q. Li, B. Wang, Y. Zhang, Q. Hong, and D. Shen, ''ScribFormer: Transformer makes CNN work better for scribble-based medical image segmentation,'' *IEEE Trans. Med. Imag.*, vol. 43, no. 6, pp. 2254–2265, Jun. 2024.

[6] K. Zhang and X. Zhuang, ''CycleMix: A holistic strategy for medical image segmentation from scribble supervision,'' in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 11646–11655.

[7] X. Luo, M. Hu, W. Liao, S. Zhai, T. Song, G. Wang, and S. Zhang, ''Scribble-supervised medical image segmentation via dual-branch network and dynamically mixed pseudo labels supervision,'' 2022, *arXiv:2203.02106*.

[8] P. Zhang, Y. Zhong, and X. Li, ''ACCL: Adversarial constrained-CNN loss for weakly supervised medical image segmentation,'' 2020, *arXiv:2005.00328*.

[9] Y. Grandvalet and Y. Bengio, ''Semi-supervised learning by entropy minimization,'' in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 17, 2004, pp. 1–8.

[10] C. Sakaridis, D. Dai, and L. Van Gool, ''ACDC: The adverse conditions dataset with correspondences for semantic driving scene understanding,'' in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10745–10755.

[11] F. Wu and X. Zhuang, ''Minimizing estimated risks on unlabeled data: A new formulation for semi-supervised medical image segmentation,'' *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 5, pp. 6021–6036, May 2023.

[12] Y. Qu, T. Lu, S. Zhang, and G. Wang, ''ScribSD+: Scribble-supervised medical image segmentation based on simultaneous multi-scale knowledge distillation and class-wise contrastive regularization,'' *Computerized Med. Imag. Graph.*, vol. 116, Sep. 2024, Art. no. 102416.

[13] Z. Li, Y. Zheng, X. Luo, D. Shan, and Q. Hong, ''ScribbleVC: Scribble-supervised medical image segmentation with vision-class embedding,'' in *Proc. 31st ACM Int. Conf. Multimedia*, Oct. 2023, pp. 3384–3393.

[14] C. Krishna, ''Accurate medical image segmentation with limited annotations,'' Ph.D. dissertation, ETH Zurich, Zurich, Switzerland, 2022. [Online]. Available: https://www.research-collection.ethz.ch/entities/publication/5aad8f54-36bd-4793-b8bd-a2c9cd1bb742

[15] G. Ian, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, ''Generative adversarial networks,'' *Commun. ACM*, vol. 63, no. 11, pp. 139–144, 2020.

[16] C. Han, L. Rundo, R. Araki, Y. Nagano, Y. Furukawa, G. Mauri, H. Nakayama, and H. Hayashi, ''Combining noise-to-image and image-to-image GANs: Brain MR image augmentation for tumor detection,'' *IEEE Access*, vol. 7, pp. 156966–156977, 2019.

[17] M. Frid-Adar, I. Diamant, E. Klang, M. Amitai, J. Goldberger, and H. Greenspan, ''GAN-based synthetic medical image augmentation for increased CNN performance in liver lesion classification,'' *Neurocomputing*, vol. 321, pp. 321–331, Dec. 2018.

[18] A. Iqbal, M. Sharif, M. Yasmin, M. Raza, and S. Aftab, ''Generative adversarial networks and its applications in the biomedical image segmentation: A comprehensive survey,'' *Int. J. Multimedia Inf. Retr.*, vol. 11, no. 3, pp. 333–368, 2022.

[19] W. Du, Y. Huo, R. Zhou, Y. Sun, S. Tang, X. Zhao, Y. Li, and G. Li, ''Consistency label-activated region generating network for weakly supervised medical image segmentation,'' *Comput. Biol. Med.*, vol. 173, May 2024, Art. no. 108380.

[20] W. Hu, H. Sheng, J. Wu, Y. Li, T. Liu, Y. Wang, and Y. Wen, ''Generative adversarial training for weakly supervised nuclei instance segmentation,'' in *Proc. IEEE Int. Conf. Syst., Man, Cybern. (SMC)*, Oct. 2020, pp. 3649–3654.

[21] G. Valvano, A. Leo, and S. A. Tsaftaris, ''Learning to segment from scribbles using multi-scale adversarial attention gates,'' *IEEE Trans. Med. Imag.*, vol. 40, no. 8, pp. 1990–2001, Aug. 2021.

[22] Y. Skandarani, P.-M. Jodoin, and A. Lalande, ''GANs for medical image synthesis: An empirical study,'' *J. Imag.*, vol. 9, no. 3, p. 69, Mar. 2023.

[23] W. Yan, Y. Wang, S. Gu, L. Huang, F. Yan, L. Xia, and Q. Tao, ''The domain shift problem of medical image segmentation and vendor-adaptation by Unet-GAN,'' in *Proc. 22nd Int. Conf. Med. Image Comput. Comput. Assist. Intervent.*, Shenzhen, China. Cham, Switzerland: Springer, Oct. 2019, pp. 623–631.

[24] A. K. Mondal, J. Dolz, and C. Desrosiers, ''Few-shot 3D multi-modal medical image segmentation using generative adversarial learning,'' 2018, *arXiv:1810.12241*.

[25] J. Mao, X. Yin, Y. Chang, and Q. Huang, ''Weakly-supervised generative adversarial networks for medical image classification,'' 2021, *arXiv:2111.14605*.

[26] Z. L. Ni, G. B. Bian, X. H. Zhou, Z. G. Hou, X. L. Xie, C. Wang, Y. J. Zhou, R. Q. Li, and Z. Li, ''Raunet: Residual attention U-Net for semantic segmentation of cataract surgical instruments,'' in *Proc. Int. Conf. Neural Inf. Process.* Cham, Switzerland: Springer, 2019, pp. 139–149.

[27] L. Zhang, W. Li, K. Bi, P. Li, L. Zhang, and H. Liu, ''FDDSeg: Unleashing the power of scribble annotation for cardiac MRI images through feature decomposition distillation,'' *IEEE J. Biomed. Health Informat.*, vol. 29, no. 1, pp. 285–296, Jan. 2025.

[28] J. Ma, J. Chen, M. Ng, R. Huang, Y. Li, C. Li, X. Yang, and A. L. Martel, ''Loss Odyssey in medical image segmentation,'' *Med. Image Anal.*, vol. 71, Jul. 2021, Art. no. 102035.

[29] M. Gong, S. Chen, Q. Chen, Y. Zeng, and Y. Zhang, ''Generative adversarial networks in medical image processing,'' *Current Pharmaceutical Design*, vol. 27, no. 15, pp. 1856–1868, Apr. 2021.

[30] O. Ronneberger, P. Fischer, and T. Brox, ''U-Net: Convolutional networks for biomedical image segmentation,'' in *Proc. Int. Conf. Med. Image Comput. Comput. Assist. Intervent.* Cham, Switzerland: Springer, 2015, pp. 234–241.

[31] D. Lin, J. Dai, J. Jia, K. He, and J. Sun, ''ScribbleSup: Scribble-supervised convolutional networks for semantic segmentation,'' in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 3159–3167.

[32] S. Zheng, S. Jayasumana, B. Romera-Paredes, V. Vineet, Z. Su, D. Du, C. Huang, and P. H. S. Torr, ''Conditional random fields as recurrent neural networks,'' in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1529–1537.

[33] B. Kim and J. C. Ye, ''Mumford–Shah loss functional for image segmentation with deep learning,'' *IEEE Trans. Image Process.*, vol. 29, pp. 1856–1866, 2020.

[34] X. Liu, Q. Yuan, Y. Gao, K. He, S. Wang, X. Tang, J. Tang, and D. Shen, ''Weakly supervised segmentation of COVID19 infection with scribble annotation on CT images,'' *Pattern Recognit.*, vol. 122, Feb. 2022, Art. no. 108341.

[35] C. F. Baumgartner, L. M. Koch, M. Pollefeys, and E. Konukoglu, ''An exploration of 2D and 3D deep learning techniques for cardiac mr image segmentation,'' in *Proc. Int. Workshop Stat. Atlases Comput. Models Heart*. Cham, Switzerland: Springer, 2017, pp. 111–119.

[36] Z. Zhou, M. M. R. Siddiquee, N. Tajbakhsh, and J. Liang, ''UNet++: Redesigning skip connections to exploit multiscale features in image segmentation,'' *IEEE Trans. Med. Imag.*, vol. 39, no. 6, pp. 1856–1867, Jun. 2020.

[37] J.-H. Kim, W. Choo, and H. O. Song, "Puzzle mix: Exploiting saliency and local statistics for optimal mixup," in *Proc. Int. Conf. Mach. Learn.*, Jun. 2020, pp. 5275–5285.

[38] H. Zhang, M. Cisse, Y. N. Dauphin, and D. Lopez-Paz, "Mixup: Beyond empirical risk minimization," 2017, *arXiv:1710.09412*.

[39] S. Yun, D. Han, S. Chun, S. J. Oh, Y. Yoo, and J. Choe, "CutMix: Regularization strategy to train strong classifiers with localizable features," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6022–6031.

[40] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7794–7803.

MANSOOR HAYAT (Member, IEEE) received the B.Sc. degree in electrical engineering from the University of Engineering and Technology, Taxila, Pakistan, in 2015, the M.Sc. degree in electrical engineering from the Institute of Southern Multan, Pakistan, in 2018, the M.B.A. degree from the National College of Business Administration and Economics, Lahore, and the Ph.D. degree in electrical engineering from Chulalongkorn University, Bangkok, Thailand, in 2024. He is currently a Postdoctoral Fellow with the Section of Neurosurgery, Department of Surgery, Rady Faculty of Health Sciences, University of Manitoba, Canada. His research interests include applying deep learning and machine learning techniques to medical imaging and video processing. In recognition of his work, he was awarded the Best Conference Paper at the 2023 IEEE Region 10 Conference (TENCON), Chiang Mai, Thailand.

SUPAVADEE ARAMVITH (Senior Member, IEEE) received the B.S. degree (Hons.) in computer science from Mahidol University, in 1993, and the M.S. and Ph.D. degrees in electrical engineering from the University of Washington, Seattle, USA, in 1996 and 2001, respectively. She joined Chulalongkorn University, in June 2001. She is currently an Associate Professor with the Department of Electrical Engineering, specializing in image and video signal processing. She has successfully advised 14 Ph.D., 30 master's, and 41 bachelor's graduates. She has published over 130 papers in international conference proceedings and journals, with four international book chapters. She has rich project management experience as the Project Leader and the former Technical Committee Chair of Thailand Government bodies in telecommunications and ICT. She is very active in the international arena with leadership positions in the global network, such as JICA Project for AUN/SEED-Net, NICT ASEAN IVO, and professional organizations, such as IEEE, IEICE, APSIPA, and ITU.